

Lab 13 Building ML Pipelines

[Start Assignment](#)

- Due Jul 25 by 11:59pm
- Points 100
- Submitting a file upload

Lab Instructions

Complete the assigned lab notebook to demonstrate hands-on understanding of the lab concepts. Follow the instructions below unless this specific lab states different requirements.

1) Required Work

You must complete the **entire notebook** and all required student sections, which may include:

- **Code sections** (examples: *“Your Code Here,” “TODO,” “Student Task,” “Complete this function”*)
- **Explanation sections** (examples: *“Explain,” “Analysis,” “Answer in your own words”*)
- **Knowledge checks** (short questions embedded in the notebook)
- **Reflection questions embedded in the notebook** (common in many labs)

Work through the notebook **in order**, run cells as you go, and review outputs carefully.

2) Where to Put Your Answers

Depending on the notebook design, you will enter your work in one or more of these places:

- **Code cells** → for required programming tasks
 - **Markdown cells** → for explanations, knowledge checks, and reflection answers (**preferred**)
 - **Short comments** → only when the notebook specifically requests it
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3) Submission Format (Choose ONE Unless the Lab Says Otherwise)

Submit your completed lab as either:

- **Option A:** the completed `.ipynb` **notebook** (with outputs visible), **OR**
- **Option B:** a **PDF export** of the notebook (code + outputs + plots readable)

If you submit as PDF

1. Run the full notebook (or all required sections).
2. Collapse/clear extremely long outputs if needed so the PDF remains readable.
3. Export: **File** → **Print** → **Save as PDF** (or **Ctrl+P** → **Save as PDF**).
4. Open the PDF and confirm it is readable and complete.

4) Reflection Requirement (Varies by Lab)

This lab will use **one** of the following reflection formats:

A) Reflection embedded in the notebook (most common)

If the notebook includes reflection questions/prompts, you must answer them **inside the notebook** (typically in Markdown cells).

✅ In this case, **no separate reflective journal is required** unless the lab explicitly says so.

B) Separate reflective journal (only if explicitly required)

If this lab specifically requires a separate journal, submit a **1–2 page reflection** as a PDF.

5) Expected Deliverables

Submit what is required for *this lab*:

1. Lab Notebook Submission (choose one):

- `L##_YourName_CourseCode.ipynb` **OR**
- `L##_YourName_CourseCode.pdf`

2. Reflection (only if required for this lab):

- Embedded responses inside the notebook **OR**
- `L##Journal_YourName_CourseCode.pdf`

(Replace `L##` with the lab number and use your course code exactly, e.g., ITAI1371.)

File Naming Penalty

Incorrectly named files will receive a -5 point deduction.

Make sure your filenames match the required convention exactly.

6) Grading Rubric (How Your Lab Will Be Assessed)

Note: Code “*technical quality*” (advanced style, optimization, cleverness) is **not** graded.

You will be graded on **correctness**, **required commenting**, and **your written understanding**.

Rubric (100 points total)

1. Completion & Correctness of Required Code (50 points)

- Required sections completed
- Code runs and produces the expected outputs/results

2. Code Comments & Clarity (15 points)

- Required comments are included where requested
- Comments explain what you did (especially in non-obvious steps)

3. Knowledge Check Questions (20 points)

- All knowledge checks answered
- Answers show understanding (not one-word or copied definitions)

4. Reflection (15 points)

- Reflection prompts completed (embedded or separate, depending on the lab)
 - Responses show real thinking: insights, challenges, connections, and takeaways
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7) Self-Check Before Submitting

- All required cells ran successfully and outputs are visible
- All **Student Task / TODO / Your Code Here** sections are completed
- Knowledge checks are answered clearly (Markdown preferred)
- Reflection is completed (embedded or separate as required)

- Submission is either `.ipynb` or readable PDF (unless told otherwise)
 - File naming matches the required convention (**-5 points if not**)
 - Work is your own (individual assignment)
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Individual Work Policy

This is an **INDIVIDUAL** assignment. You may use course materials, official documentation, and general programming resources. Your code, explanations, and reflections must be your own.

Need Help?

If you have questions while working on the lab, **do not hesitate to contact me** via **em**